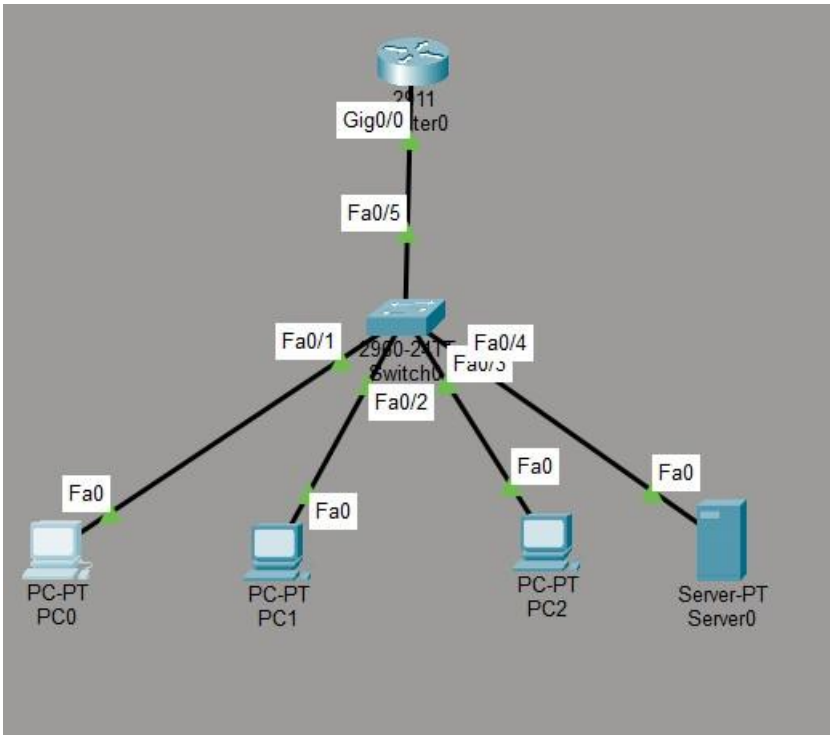
Section C — Mini Capstone Project

Qus. Mini Project: TechLaunch Startup Network Build & Fault Diagnosis
Objective: Build a complete two-VLAN network for a startup, then diagnose and resolve a Router-ona-Stick failure following a power outage — combining VLAN segmentation, trunking, inter-VLAN routing, and structured fault-finding into one exercise.



IP plan

Device	VLAN	IP Address	Gateway
PC0 Marketing	10	192.168.10.10/24	192.168.10.1
PC1 Marketing	10	192.168.10.11/24	192.168.10.1
PC2 Engineering	20	192.168.20.10/24	192.168.20.1
File Server	20	192.168.20.20/24	192.168.20.1
R1 G0/0.10	10	192.168.10.1/24	—
R1 G0/0.20	20	192.168.20.1/24	—

Configure the switch

```
Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name Marketing
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name Engineering
Switch(config-vlan)#exit
```

Assign ports

```
Switch(config)#int range f0/1-2
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#exit
Switch(config)#int range f0/3-4
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 20
Switch(config-if-range)#exit
```

Configure trunk

```
Switch(config)#int f0/5
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk allowed vlan 10,20
Switch(config-if)#exit
Switch(config)#do wr
Building configuration...
[OK]
Switch(config)#
```

Configure Router

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int g0/0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#int g0/0.10
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.10, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up

Router(config-subif)#en
Router(config-subif)#encapsulation dot1q 10
Router(config-subif)#ip address 192.168.10.1 255.255.255.0
Router(config-subif)#exit
Router(config)#do wr
Building configuration...
[OK]
Router(config)#int g0/0.20
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.20, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up

Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 192.168.20.1 255.255.255.0
Router(config-subif)#exit
Router(config)#do wr
Building configuration...
[OK]
Router(config)#int g0/0.20
Router(config-subif)#no encapsulation dot1Q 20
Router(config-subif)#exit
Router(config)#do wr
Building configuration...
[OK]
Router(config)#
```

Test the working network

```
C:\>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time=19ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time=4ms TTL=128

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 19ms, Average = 5ms
```

Engineering gateway troubleshooting

```
C:\>ping 192.168.20.2

Pinging 192.168.20.2 with 32 bytes of data:

Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Request timed out.
Reply from 192.168.10.1: Destination host unreachable.

Ping statistics for 192.168.20.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Switch verification commands

```
Switch#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/5       on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/5       10,20

Port        Vlans allowed and active in management domain
Fa0/5       10,20

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/5       10,20

Switch#
```

```
Switch#show interfaces gigabitEthernet0/1 switchport
Name: Gig0/1
Switchport: Enabled
Administrative Mode: dynamic auto
Operational Mode: down
Administrative Trunking Encapsulation: dot1q
Operational Trunking Encapsulation: native
Negotiation of Trunking: On
Access Mode VLAN: 1 (default)
Trunking Native Mode VLAN: 1 (default)
Voice VLAN: none
Administrative private-vlan host-association: none
Administrative private-vlan mapping: none
Administrative private-vlan trunk native VLAN: none
Administrative private-vlan trunk encapsulation: dot1q
Administrative private-vlan trunk normal VLANs: none
Administrative private-vlan trunk private VLANs: none
Operational private-vlan: none
Trunking VLANs Enabled: All
Pruning VLANs Enabled: 2-1001
Capture Mode Disabled
Capture VLANs Allowed: ALL
Protected: false
Unknown unicast blocked: disabled
Unknown multicast blocked: disabled
Appliance trust: none


Switch#
```

Router verification

```
interface GigabitEthernet0/0
 no ip address
 duplex auto
 speed auto
!
interface GigabitEthernet0/0.10
 encapsulation dot1Q 10
 ip address 192.168.10.1 255.255.255.0
!
interface GigabitEthernet0/0.20
 no ip address
!
interface GigabitEthernet0/1
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface GigabitEthernet0/2
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface Vlan1
 no ip address
 shutdown
!
ip classless
!
ip flow-export version 9
!
```

Root cause identified.

The VLAN 20 Router-on-a-Stick sub-interface is missing its 802.1Q encapsulation.

Fix the fault

```
Router(config)#int g0/0.20
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.20, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up

Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 192.168.20.1 255.255.255.0
Router(config-subif)#exit
Router(config)#do wr
Building configuration...
[OK]
```

Now test again

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.2

Pinging 192.168.20.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.20.2: bytes=32 time=1ms TTL=127
Reply from 192.168.20.2: bytes=32 time=6ms TTL=127
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.20.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 6ms, Average = 2ms
```

Inter-VLAN routing restored.

Configuration:
VLAN 10 and VLAN 20 were created on the switch and access ports were assigned to their respective departments. The switch-router connection was configured as an 802.1Q trunk. Router subinterfaces G0/0.10 and G0/0.20 were configured as the default gateways for the two VLANs.

Fault:
After simulating a router post-power-outage failure, Marketing devices could communicate with devices in their own VLAN but could not reach the Engineering file server.

Root Cause:
The encapsulation dot1Q 20 configuration was missing from router sub-interface G0/0.20.
Therefore, VLAN 20 traffic could not be correctly processed by the Router-on-a-Stick configuration.

After applying the configuration, Marketing could successfully ping the Engineering gateway and file server, confirming that inter-VLAN routing had been restored.